

Mathematics MCQ Test — Answer Key

Companion to `advanced-math-mcq.md`.

Q1 — B

Use $\log a + \log b = \log(ab)$:

$$\log_2 ((x-1)(x+1)) = 3 \implies (x-1)(x+1) = 2^3 = 8.$$

So $x^2 - 1 = 8$, hence $x^2 = 9$ and $x = 3$ (domain requires $x > 1$).

Check: $\log_2 2 + \log_2 4 = 1 + 2 = 3$.

Q2 — B

First term $a = 15$, common ratio $r = \frac{1}{3}$ with $|r| < 1$, so

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} = \frac{15}{1-\frac{1}{3}} = \frac{15}{\frac{2}{3}} = \frac{45}{2} = 22.5.$$

Q3 — D

Square both sides:

$$(\sin \theta + \cos \theta)^2 = \frac{1}{9} \implies \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = \frac{1}{9}.$$

So $1 + \sin 2\theta = \frac{1}{9}$, hence $\sin 2\theta = -\frac{8}{9}$.

Q4 — A

Double-angle formula: $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$ with $\tan \alpha = \frac{1}{3}$:

$$\frac{2 \cdot \frac{1}{3}}{1 - \left(\frac{1}{3}\right)^2} = \frac{\frac{2}{3}}{\frac{8}{9}} = \frac{2}{3} \cdot \frac{9}{8} = \frac{3}{4}.$$

Q5 — A

Set $y = \frac{x-2}{x+1}$ and solve for x :

$$y(x+1) = x-2 \implies yx + y = x-2 \implies y+2 = x(1-y).$$

Thus $x = \frac{y+2}{1-y}$. Swapping variables gives $f^{-1}(x) = \frac{x+2}{1-x}$.

Q6 — B

Taylor: $e^{3x} = 1 + 3x + \frac{(3x)^2}{2!} + O(x^3) = 1 + 3x + \frac{9x^2}{2} + O(x^3)$.

Numerator is $\frac{9x^2}{2} + O(x^3)$, so the limit is $\frac{9}{2}$.

(Alternatively, apply L'Hôpital twice to the $0/0$ form.)

Q7 — B

For $x < 0$, $|x| = -x$, so $\frac{|x|}{x} = \frac{-x}{x} = -1$.

The left-hand limit is therefore -1 .

Q8 — A

Taylor: $\ln(1+x) = x - \frac{x^2}{2} + O(x^3)$.

So $\ln(1+x) - x = -\frac{x^2}{2} + O(x^3)$, and the limit is $-\frac{1}{2}$.

Q9 — C

Divide numerator and denominator by x^3 :

$$\frac{2 + \frac{1}{x^2}}{1 - \frac{1}{x^3}} \rightarrow \frac{2}{1} = 2 \quad \text{as } x \rightarrow \infty.$$

Q10 — B

For $x \neq 3$, $\frac{x^2 - 9}{x - 3} = x + 3$.

Continuity at $x = 3$ needs $k = \lim_{x \rightarrow 3} (x + 3) = 6$.

Q11 — C

By the Intermediate Value Theorem: if f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then f has a root in (a, b) .

Q12 — A

Chain rule: $\frac{dy}{dx} = \cos(x^2 e^x) \cdot \frac{d}{dx}(x^2 e^x)$.

Product rule: $\frac{d}{dx}(x^2 e^x) = 2x e^x + x^2 e^x = (2x + x^2) e^x$.

Q13 — C

Differentiate implicitly: $2x + 2y y' = 0$, so $y' = -\frac{x}{y}$ (where $y \neq 0$).

Q14 — A

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, so the coefficient of x^3 is $\frac{1}{3!} = \frac{1}{6}$.

Q15 — B

$f'(x) = 3x^2 - 3 = 3(x^2 - 1)$, so critical points are $x = \pm 1$.

$f''(x) = 6x$: $f''(-1) < 0$ (local max), $f''(1) > 0$ (local min).

Q16 — B

Integration by parts with $u = x$, $dv = e^x dx$:

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C = e^x(x - 1) + C.$$

Q17 — A

Use $\sin^2 x = \frac{1 - \cos 2x}{2}$:

$$\int \sin^2 x \, dx = \int \frac{1 - \cos 2x}{2} \, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

Q18 — A

Partial fractions: $\frac{1}{x^2 - 1} = \frac{1/2}{x - 1} - \frac{1/2}{x + 1}$.

Integrating gives $\frac{1}{2} \ln |x - 1| - \frac{1}{2} \ln |x + 1| + C = \frac{1}{2} \ln \left| \frac{x - 1}{x + 1} \right| + C$.

Q19 — A

Fundamental Theorem of Calculus with chain rule: if $F(x) = \int_a^{g(x)} f(t) \, dt$, then $F'(x) = f(g(x)) g'(x)$.

Here $F'(x) = \cos(x^2) \cdot 2x$.

Q20 — B

$$\int_1^\infty x^{-2} \, dx = \lim_{b \rightarrow \infty} \left[-\frac{1}{x} \right]_1^b = \lim_{b \rightarrow \infty} \left(-\frac{1}{b} + 1 \right) = 1.$$

Q21 — B

Use $\cos^2 x = \frac{1 + \cos 2x}{2}$:

$$\int_0^{\pi/2} \cos^2 x \, dx = \int_0^{\pi/2} \frac{1 + \cos 2x}{2} \, dx = \left[\frac{x}{2} + \frac{\sin 2x}{4} \right]_0^{\pi/2} = \frac{\pi}{4}.$$

Q22 — A

Curves meet when $x = x^2$, so $x = 0$ or $x = 1$. For $x \in [0, 1]$, $x \geq x^2$, so

$$\text{Area} = \int_0^1 (x - x^2) \, dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Q23 — A

Separable: $\frac{dy}{y} = 3 dx$ gives $\ln |y| = 3x + C$, so $y = Ae^{3x}$.

$y(0) = 4$ implies $A = 4$, hence $y = 4e^{3x}$.

Q24 — C

A is upper triangular, so $\det(A)$ is the product of the diagonal entries:

$$\det(A) = 1 \cdot 4 \cdot 6 = 24.$$

Q25 — B

A is upper triangular, so the eigenvalues are the diagonal entries 4 and 2.

Q26 — C

$\det(A) = 0$ means A is singular (not invertible), so the homogeneous system $Ax = 0$ has a nontrivial solution.

Q27 — A

$$\det(A) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

$$A^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}.$$

Q28 — B

Row 2 is twice row 1, so those two rows are dependent. Row 3 is not a multiple of row 1, so the rank is 2.

Q29 — B

$Ax = 0$ gives $x + 2y = 0$, so $x = -2y$. Solutions are multiples of $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$, equivalently spanned by $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

Q30 — C

For an $n \times n$ matrix, $\det(cA) = c^n \det(A)$.

Here $n = 3$ and $c = 2$, so $\det(2A) = 2^3 \cdot 5 = 8 \cdot 5 = 40$.

Quick Reference

Q	Answer	Q	Answer	Q	Answer
1	B	11	C	21	B
2	B	12	A	22	A
3	D	13	C	23	A
4	A	14	A	24	C
5	A	15	B	25	B
6	B	16	B	26	C
7	B	17	A	27	A
8	A	18	A	28	B
9	C	19	A	29	B
10	B	20	B	30	C